Long Short-Term Memory Networks for Fast Optical Signal Identification in the Human Visual Cortex for Brain Computer Interface Applications

David Perpetuini[1,*], Giulia Rocco[2], Elizabeth Fear[2], Sara Pomante[2], Lucie Chalet[2], Francesca Graziano[2], Ettore Valeri[1], Manuela Carriero[2], Cosimo Del Gratta[2], Mauro Gianni Perrucci[2], Richard Wise[2] and Antonio Maria Chiarelli[2]

[1] Department of Engineering and Geology, University "Gabriele D'Annunzio" of Chieti-Pescara, 65127 Pescara, Italy
[2] Department of Neuroscience, Imaging and Clinical Sciences, "G. D'Annunzio" University of Chieti-Pescara, 66100, Chieti, Italy

david.perpetuini@unich.it

Abstract. A brain-computer interface (BCI) is a technology that translates brain signals into commands to interact with devices. To this aim, portable neuroimaging modalities, such as functional near-infrared spectroscopy (fNIRS), are particularly appropriate. It is worth noticing that optical signals in the near infrared spectral range can provide information also regarding rapid neural optical properties modulations associated with their depolarization, which are known as fast optical signals (FOS). FOS are characterized by a high spatiotemporal resolution, but also by a poor signal-to-noise ratio, restricting their applicability in BCIs. In order to enhance the feasibility of employing FOS in BCI, approaches based on artificial intelligence (AI) could be beneficial given its capability to identify structures in the data. In this study, an AI approach was employed for FOS identification. Specifically, FOS were obtained using a continuous-wave near-infrared system from the visual cortex in response to a visual stimlus. A photon count, using Direct Current (DC) light intensity at 830 nm, was integrated with an AI methodology for the rapid assessment of visual cortex brain activity. Particularly, a Long Short-Term Memory (LSTM) model was used at the subject level to the optical data, using its capacity to describe temporal dependencies across several acquisition channels for the categorization of visual cortex activity. An average accuracy of 62.8% accuracy was achieved in distinguishing cortical activity from rest across participants. This aproach is a first endeavor to detect visual cortex activity using FOS, facilitating the development of FOS-based BCI systems.

Keywords: near infrared optical imaging, event-related optical signals (EROS), diffuse optical tomography , Deep Learning, Visual Cortex

1 Introduction

Brain-computer interfaces (BCIs) can facilitate direct communication between the human brain and external devices by interpreting neural activity [1]. In this context, non-invasive and portable neuroimaging techniques are preferred. Specifically, the electroencephalography (EEG) is able to measure the electrical activity across the scalp, providing high temporal resolution, hence allowing for real-time processing of brain activity, but coupled with low spatial resolution [2]. In contrast, functional near-infrared spectroscopy (fNIRS) can capture hemodynamic oscillations associated with the brain activity, providing a higher spatial resolution with respect to EEG but also a lower temporal resolution [3]. Importantly, near infrared imaging can provide information also regarding the modification of the neural optical properties related with the neurons depolarization, which are known as event-related optical signals (EROS) or fast optical signals [4]. The evaluation of FOS can further enhance the capabilities of BCI by offering high temporal resolution in capturing brain dynamics linked to task performance coupled with a high spatial resolution [5]. However, it should be highlight that FOS recordings require a precise localization of the optical probes, since these signals are confined to certain temporal and spatial dimensions. Moreover, FOS are characterized by a low signal to noise ratio (SNR), thus requiring measuring several trials repetitions and averaging the responses. To improve the detectability of FOS, many data analysis methodologies have been suggested, including spectral evaluation [6], Independent Component Analysis [7], and approaches based on the Generalized Linear Model [8]. In addition, it was demonstrated that acquiring FOS using wavelengths longer than 800 nm can enhance SNR because to their reduced absorption properties [9].

Importantly, it is worth highlighting that FOS combined with machine learning (ML) methods could represent an effective approach for BCI applications. In fact, the elevated FOS' temporal resolution may provide higher information transfer rate (ITR) with respect to fNIRS, which is a crucial aspect in BCI, and ML methods could facilitate the FOS' data analysis, providing a classification of the brain activity [10]. However, it should be noted that FOS has seldom been used in combination with data-driven ML methodologies for classifying cortical activity and achieving effective categorization of brain signals. For instance, Chiou et al., (2024) applied a convolutional neural network for single-trial classification of FOS related to motor activity, reaching an accuracy of around 63% [11], whereas Perpetuini et al., (2023) proposed a method combining wavelet coherence and support vector machine for retinotopy mapping, achieving an accuracy of around 63% and ITR of 6 bpm [12].

In this context, since ML and especially deep learning (DL) approaches are very successful in categorizing brain signals [13], it could be worth investing in the feasibility to employ DL approaches for brain signals categorization based on FOS. Specifically, Long Short-Term Memory (LSTM) [14], a specialized kind of recurrent neural network (RNN) designed to learn and retain long-term relationships in sequential data, has shown efficacy in categorizing brain activity, but it has not been used for recognizing FOS. This research aims to examine the efficacy of integrating FOS with LSTM for classifying visual cortical activity based on FOS for relevant BCI applications.

2 Materials and Methods

2.1 Participants

Five healthy volunteers (ages 24–38 years, mean age 28 years, four males) participated in the trial. Participants were right-handed, had normal or corrected eyesight and hearing, and self-reported excellent health, free from drugs. All participants were apprised of the research methodology and furnished their informed consent. The research received approval from the local Ethical Committee and was conducted in accordance with the ethical principles of the Helsinki Declaration.

2.2 Visual Stimulation

The visual stimulus was a black and white rectangular checkerboard flickering at 2 Hz. In the center of the screen was presented a fixation cross that participants had to look at. The stimulation was composed of 10 blocks of 30s separated by periods of 60s of rest. This paradigm provided 600 trials for each subject. Stimuli were presented on a screen through the E-Prime 3.0 software (Psychology Software Tools, Pittsburgh, PA).

2.3 Optical Imaging Recordings

Optical data were acquired using the Borealis system (NIRx, Germany), a dual-wavelength (i.e., 785 nm and 830 nm) continuous-wave device designed for high-density optical imaging. The optodes were arranged to deliver 114 measurement channels on the visual cortex at a sampling frequency of 15 Hz, sufficient for FOS detection. The montage is reported in Fig. 1.

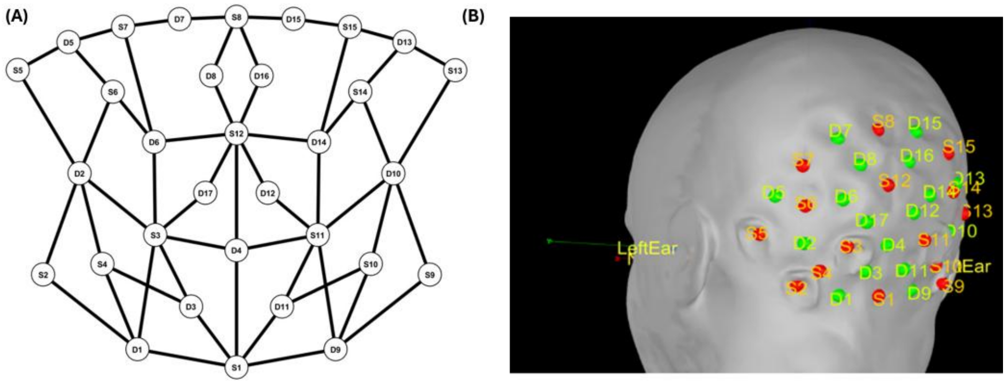

Fig. 1. (A) Representation of the optodes montage. The black lines represent only the shorter channels: all the combinations between sources and detectors delivered the 114 measuring channels. (B) Optodes montage projected on a scalp. The red dots represent the sources, whereas the green dots are the detectors.

2.4 FOS data analysis

The DC intensity signal at 830 nm wavelength was employed for the analysis. A zero-phase lag digital high-pass filter with a 2 Hz cutoff frequency was employed to the DC signals to remove the parts of the signasl associated with the hemodynamic. The DC variations were assessed by transforming the signals into optical densities. Signals obtained from all accessible channels have been fed into the classifier.

2.5 LSTM approach and dataset preparation

The dataset was randomly split into training (70%) and testing (30%) stratified subsets to ensure a proper evaluation of model performance. The two classes in the dataset were balanced. An LSTM architecture was implemented in Matlab 2024b for binary classification of visual cortex activity (i.e., the activation and non-activation) based on multichannel optical short time series (0.5s). Preprocessing involved the per-event, per-channel management of absent samples through linear interpolation along the temporal axis and z-score normalization over time. The architecture consisted of an input layer (input size = number of channels), a dense projection layer with ReLU activation, and a dropout rate of 0.20, followed by a single LSTM layer with 128 hidden units in "last" output mode, an additional dropout layer with a rate of 0.30, and a fully connected layer that maps to the two classes using softmax and cross-entropy loss. Within the training set, a 5-fold cross-validation was implemented to reduce possible overfitting effects. Training used Adam (initial learning rate 10^{-3}), gradient clipping (threshold = 1), mini-batches of 5, 30 epochs, shuffling each epoch, and on-the-fly validation on the held-out set per epoch. The test performance was measured by evaluating the confusion matrix, the accuracy and the ITR.

3 Results

Figure 2 reports the confusion matrices associated with the classification task for each participant.

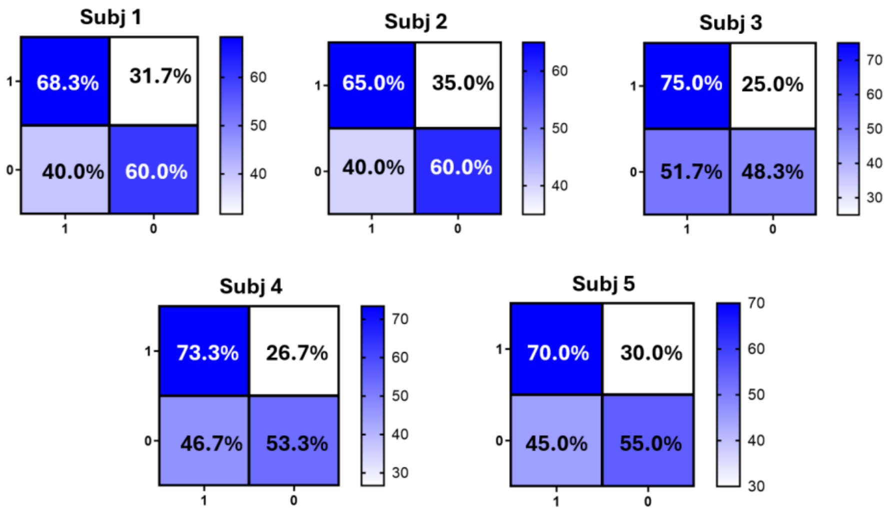

Fig. 1 Confusion matrices of the classification performance delivered by the LSTM approach for each participant.

Table 1 reports the classification accuracies for all the participants and the associated ITR. The best classification was obtained for the first participant, reaching an accuracy of 64.17% and an ITR of 6.88.

Table 1. Performance of the models is expressed as accuracy and ITR

Subject	Accuracy (%)	ITR (bpm)
Subj 1	64.17	6.88
Subj 2	62.50	5.47
Subj 3	61.67	4.92
Subj 4	63.30	6.00
Subj 5	62.50	4.92

4 Discussion

This research examined a FOS-based framework for BCI applications. Specifically, an LSTM model for binary categorization of visual cortical activity (i.e., activation vs non-activation) using multichannel optical short time series (0.5s) was developed. Only the 830 nm wavelength FOS was used, since evidence from the literature indicates that FOS exhibits heightened sensitivity to brain activity at greater wavelengths [15]. This is due to the augmented spectrum absorption at shorter wavelengths, leading to a diminished number of photons reaching the detector and a reduced SNR [16]. The results showed an average accuracy across participants of 62.8% and an ITR of 5.6 bpm. However, it is important to highlight that the average categorization accuracy fell short of the 70% threshold of the minimum standard for efficient BCI communication [17]. This research evaluated individual trials for FOS without averaging, marking an advancement above canonical FOS investigations.

Approaches of signal enhancement, signal classification, and noise mitigation may be used to enhance the efficacy of the procedure. In addition, information may be augmented by integrating optical data with other rapid brain monitoring techniques, such as EEG.

Importantly, it should be noted that physiological noise can hinder brain state

classification, resulting in a wrong classification [18]. However, FOS spectral bandwidth operates at a higher frequency range with respect to the majority of the physiological processes, resulting robust against physiological artefacts. Moreover, the configuration employed in this investigation, characterized by different inter-optode distances, results in several penetration depths that enable the isolation of effects on deep brain layers, therefore detaching them from scalp layers [19].
It is worth to highlight that developing methods for BCI based on FOS can enhance the spatiotemporal resolution of the technology, providing a higher ITR compared to fNIRS for real-time applications. Nonetheless, FOS have decreased SNR compared to fNIRS, thus impacting its effectiveness for real-time evaluations. Hence, in order to improve the performance of the method, additional experiments with an increased number of participants and trials should be conducted. Moreover, different AI approaches should be tested to enhance the accuracy and the ITR delivered.
Importantly, the novelty of this work resides in its capacity to categorize visual cortex activity using FOS without the need for trial averaging within a 0.5s time frame. This outcome may promote the use of optical signals in real-time BCI applications and foster the employment of more sophisticated DL approaches able to explore high-order non-linearities in bigger datasets to enhance classification performance.

5 Conclusions

This study presents a FOS-based method for classifying visual cortex activity useful for BCI applications. By implementing a LSTM architecture, the proposed approach achieved an average classification accuracy of 62.8% and a ITR of 5.6 bpm, with a response time of only 0.5s. Although preliminary, these findings demonstrate the feasibility of using FOS for single-trial visual cortex classification. However, further studies are indeed necessary to enlarge the study sample, in order to further test the generalizability of the results and further AI based approaches should be tested to increase the accuracy performance, in order to make FOS suitable for BCI systems.

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